

ANTIMICROBIAL SUSCEPTIBILITY PATTERNS OF GRAM-NEGATIVE UROPATHOGENS IN SOUTH WEST NIGERIA

1. SOURCE PUBLICATION

The dataset was sourced from Mendeley Data

Otaigbe, Idemudia; ELIKWU, CHARLES; Ebeigbe, Ejime (2023), "Dataset on Antibiotic susceptibility patterns of Gram negative uropathogens from patients in a tertiary hospital in South west Nigeria", Mendeley Data, V1, doi: 10.17632/nd4w9wnmzc.1.

The dataset is associated with the peer-reviewed journal article (Otaigbe et al., 2023)

2. INTRODUCTION

Urinary tract infections (UTIs) are among the most common bacterial infections encountered in clinical practice. UTIs are most often caused by members of the family Enterobacteriaceae. Antimicrobial resistance among uropathogens has increased substantially, compromising commonly used therapies. Therefore, there is a need to assess antimicrobial susceptibility patterns of urinary pathogens to provide evidence to support appropriate empirical antibiotic selection, strengthen antimicrobial stewardship efforts, and contribute to ongoing surveillance of antimicrobial resistance.

The analysis aims to determine the distribution and antimicrobial susceptibility patterns of Gram-negatives

2.1 Objectives for the analysis

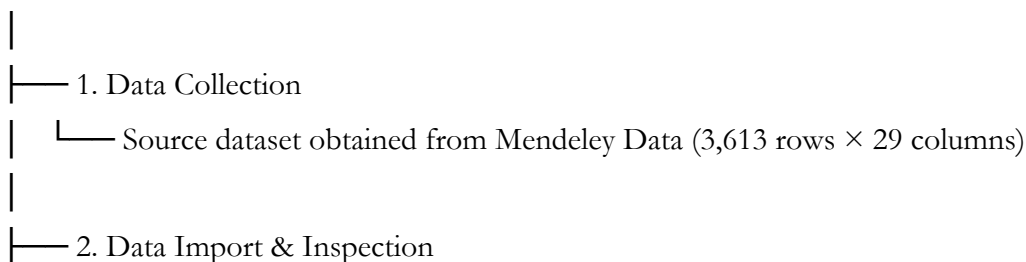
- a. To identify the most prevalent Gram-negative species isolated from patients
- b. To quantify resistance rates across major antimicrobial classes
- c. To determine the prevalence of multidrug resistance (MDR) patterns among the isolates

3. METHODOLOGY

3.1 Workflow

The workflow used for the analysis is as follows:

Project Workflow



- | | — Load dataset into R
- | | — Initial structure check (skimr, glimpse)
- |
- | — 3. Data Cleaning
 - | | — Merge S/I/R columns per antibiotic (29 → 13 columns)
 - | | — Fix S/N column data type (date → text) — A1096, A2038, A2052
 - | | — Detect & remove true duplicates (6 rows)
 - | | — Correct typographical S/N errors (5 rows)
- |
- | — 4. Data Transformation
 - | | — Standardize CULTURE and SPECIMEN columns
 - | | — Remove empty column
 - | | — Standardize column names
- |
- | — 5. Exploratory Data Analysis
 - | | — Demographic characteristics (age, gender, culture result)
 - | | — Etiological spectrum of isolated uropathogens
- |
- | — 6. Antimicrobial Susceptibility Analysis
 - | | — Overall resistance rates per antibiotic
 - | | — Resistance profiles stratified by organism
- |
- | — 7. Multidrug Resistance (MDR) Analysis
 - | | — Classify isolates by resistance count (single/dual/multi)
 - | | — Identify common co-resistance patterns
- |
- | — 8. Reporting & Visualization
 - | | — Generate charts (ggplot2)
 - | | — Generate summary tables
 - | | — Compile final report

3.2 Tools used

Packages used for the analysis include: tidyverse, janitor, readxl, skimr, stringr, scales, ggplot2

3.3 Dataset description

This original dataset contains 3613 rows and 29 columns. Each row represents an antibiotic susceptibility test result for a bacterial isolate obtained from a patient sample. Key variables include sex, ward/clinic, specimen, culture, organisms, antibiotics tested, antimicrobial susceptibility results.

3.4 Data cleaning and tidying

3.4.1 Changes made to the original dataset

The original dataset had 29 columns with each antibiotic having a separate column for susceptible(S), intermediate (I), and resistant(R). These three columns were merged into one column for each antibiotic reducing the number of columns to 13 columns.

After uploading the dataset into R, the output indicated that three of the serial numbers in column S/N appeared as dates instead of numbers. The correction was made by converting the content of the S/N column into text. The S/N affected include A1096, A2038,A2052

After screening for duplicates, 11 of the rows were reported to be duplicates. Out of the 11 duplicates, six(6) of them were confirmed to be true duplicates and were removed, while 5(five) of them were confirmed to be typographical errors in isolate serial numbers and were corrected.

Variable	Original	Corrected	Reason
S/N	487(at A848)	847	Typographical error
S/N	649 (at A550)	549	Typographical error
S/N	738 (at A639)	638	Typographical error
S/N	2593(at A2583)	2583	Typographical error
S/N	3518(at A3492)	2583	Typographical error

3.4.2 Variable transformations

After viewing the content of each column, the 'CULTURE' and 'SPECIMEN' columns were standardized. An empty column was identified and removed. The column names were all standardized.

4. RESULTS AND DATA PRESENTATION

4.1 Demographic Characteristics of the Sample

Table 1: Specimen characteristics

Culture	n	Percentage
Negative	2799	77.62
Positive	807	22.38

Of the 3606 samples, 807(22.38%) were positive cultures, while 2799(77.62%) of them were negative cultures. Organism identification and subsequent were applied to culture-positive specimens; culture-negative specimens had no organism recorded.

4.2 Etiological Spectrum of Isolated Uropathogens

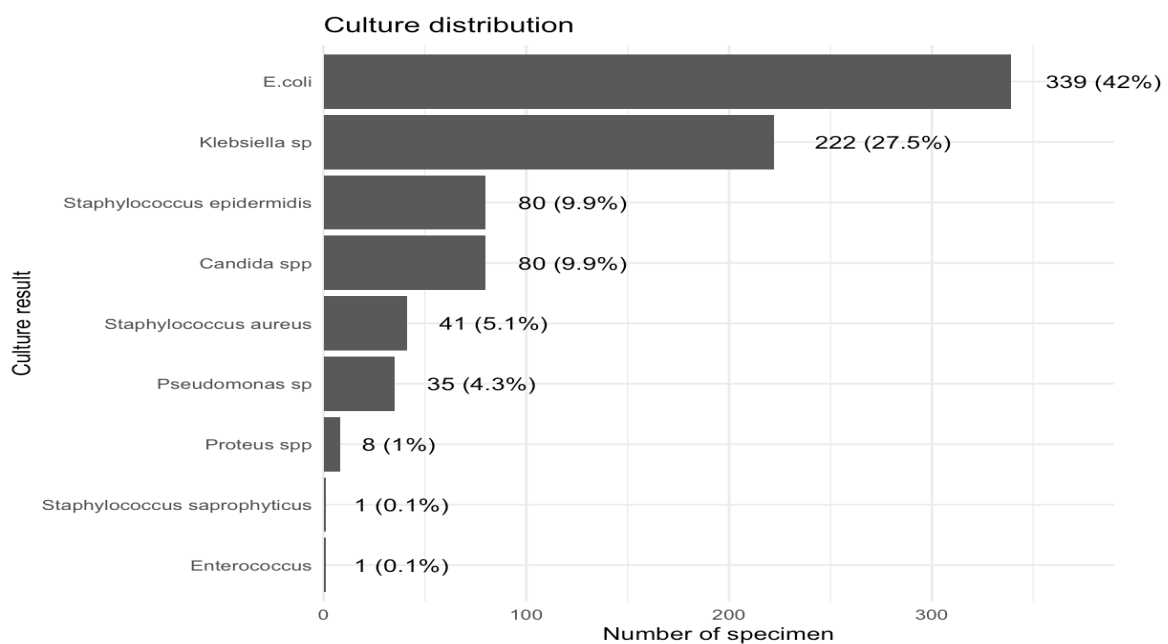


Figure 1: Distribution of microorganisms isolated

E.coli was the most prevalent uropathogen, 339(42%), followed by *Klebsiella sp.* 222(27.5%), with *Staphylococcus saprophyticus* 1(0.1%) and *Enterococcus* 1 (0.1%) being the least. These findings corroborates with that of Mareş et al., (2023) , who reported *E .coli* (54.53%) as the most frequent microbe, followed by *Klebsiella spp.* (16.54%)

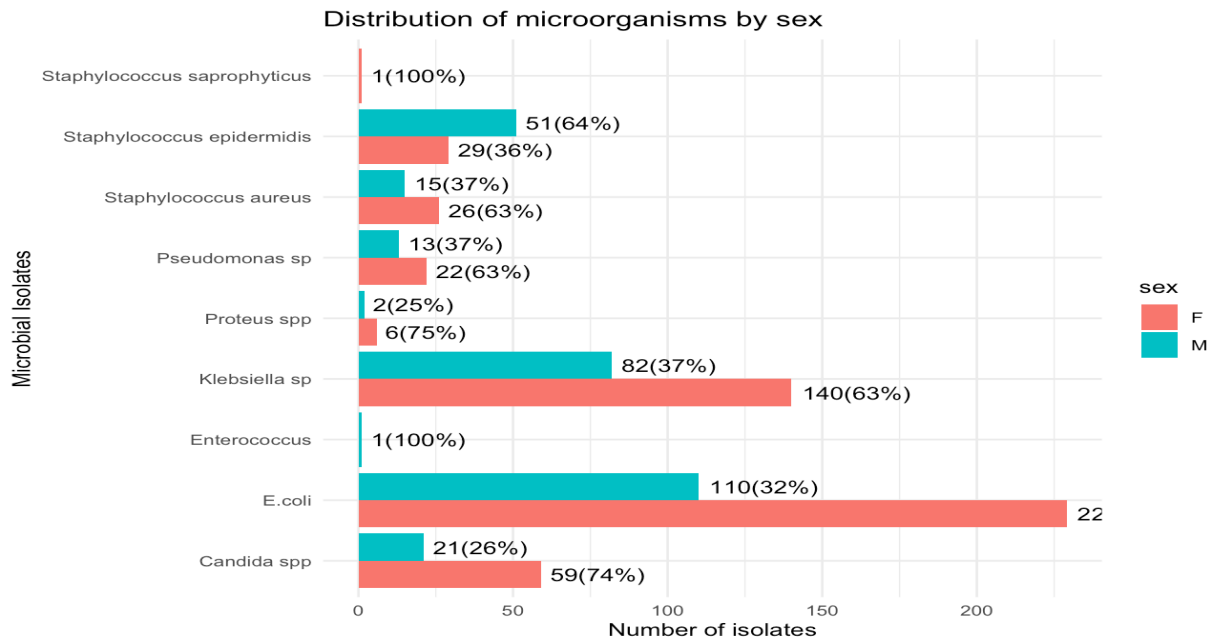


Figure 2: Distribution of microorganisms by sex

The distribution of microorganisms by sex revealed that females accounted for the majority of isolates across most organisms, with *E. coli* and *Klebsiella spp.* representing the most frequently isolated microorganisms in both sexes. *Staphylococcus saprophyticus* and *Enterococcus* were exclusively identified in males, each with a single isolate.

4.3 Antimicrobial Susceptibility Profiles

Antimicrobial susceptibility testing was performed for the Gram-negative isolates. Six antimicrobial agents were tested across these isolates, while ceftriaxone susceptibility testing was performed only for the 399 *E.coli* isolates. Gram-positive isolates and *Candida spp.* did not undergo AST using the antimicrobial panel included in this dataset. Ceftriaxone susceptibility testing was performed for the 339 *E. coli* isolates and was therefore analysed separately using the 339 *E.coli* isolates as the denominator.

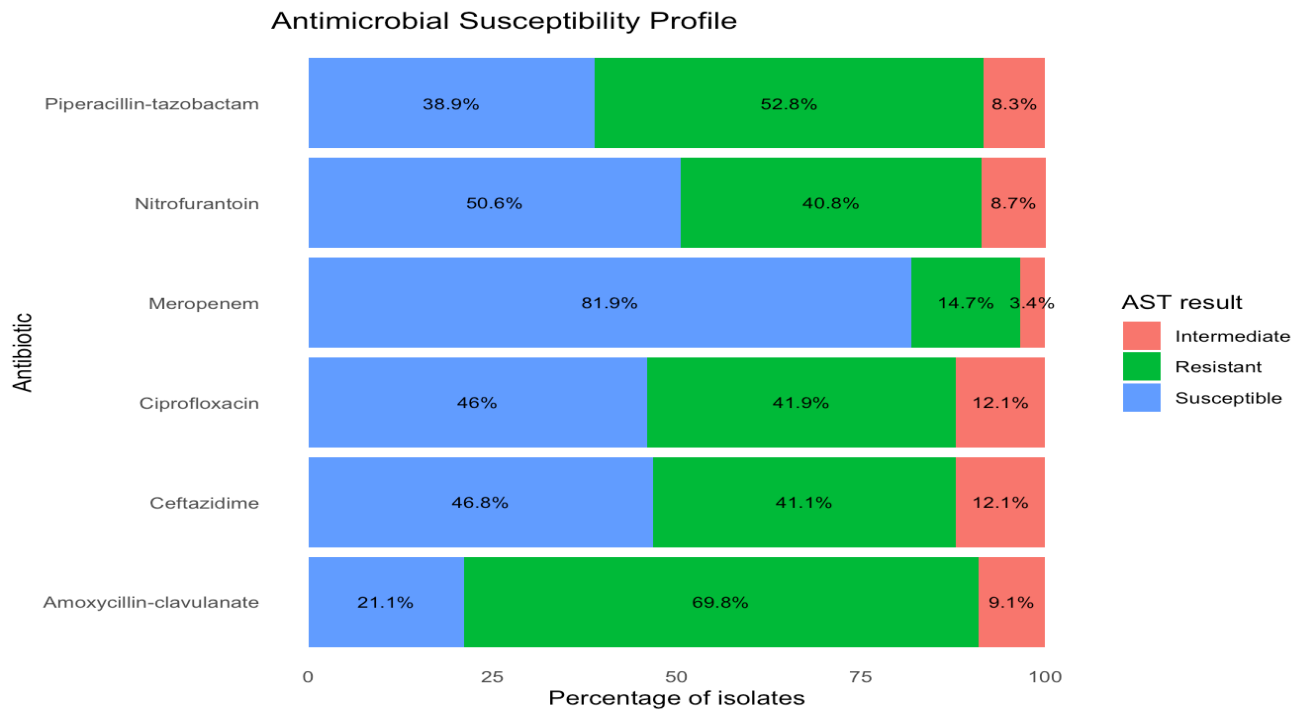


Figure 3: Antimicrobial susceptibility profile of the test antibiotics

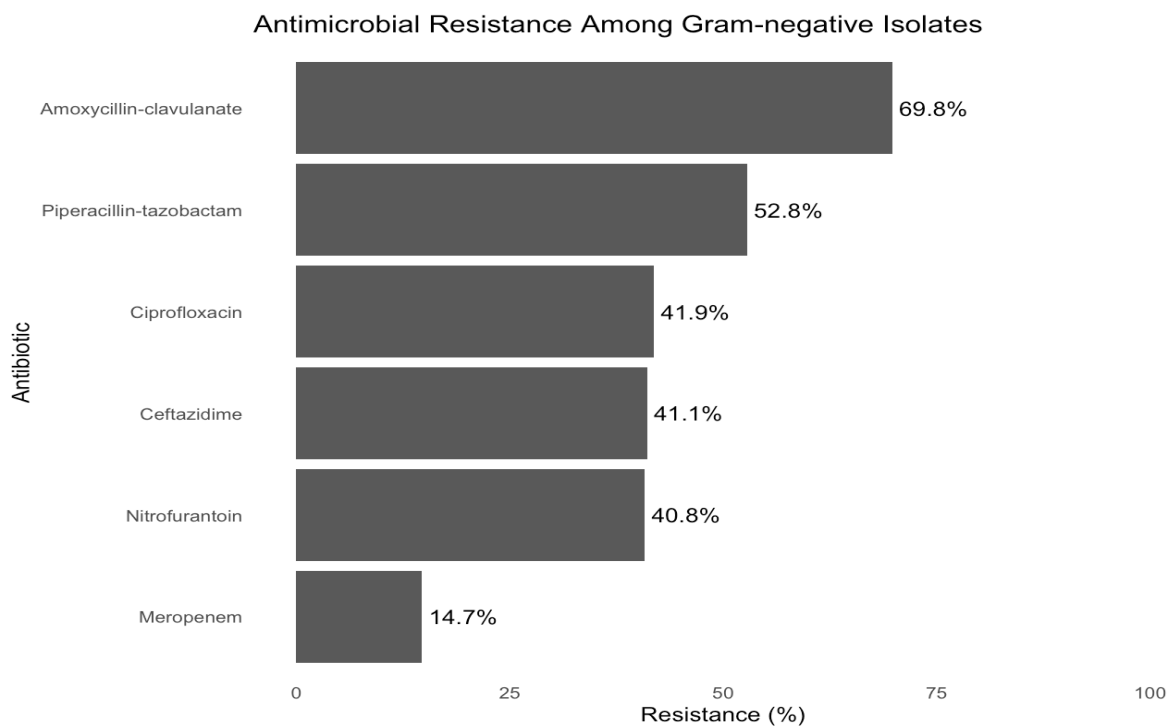


Figure 4: Resistance percentage by antibiotic

Amoxicillin-clavulanate showed the highest resistance (69.8%), while meropenem showed the highest susceptibility (81.9%). This could result from the frequency of their use in the treatment

of UTIs. Moderate susceptibility was observed for nitrofurantoin (50.6%) and ciprofloxacin (46%), whereas substantial resistance was also seen with piperacillin-tazobactam (52.8%) and ceftazidime (41.1%).

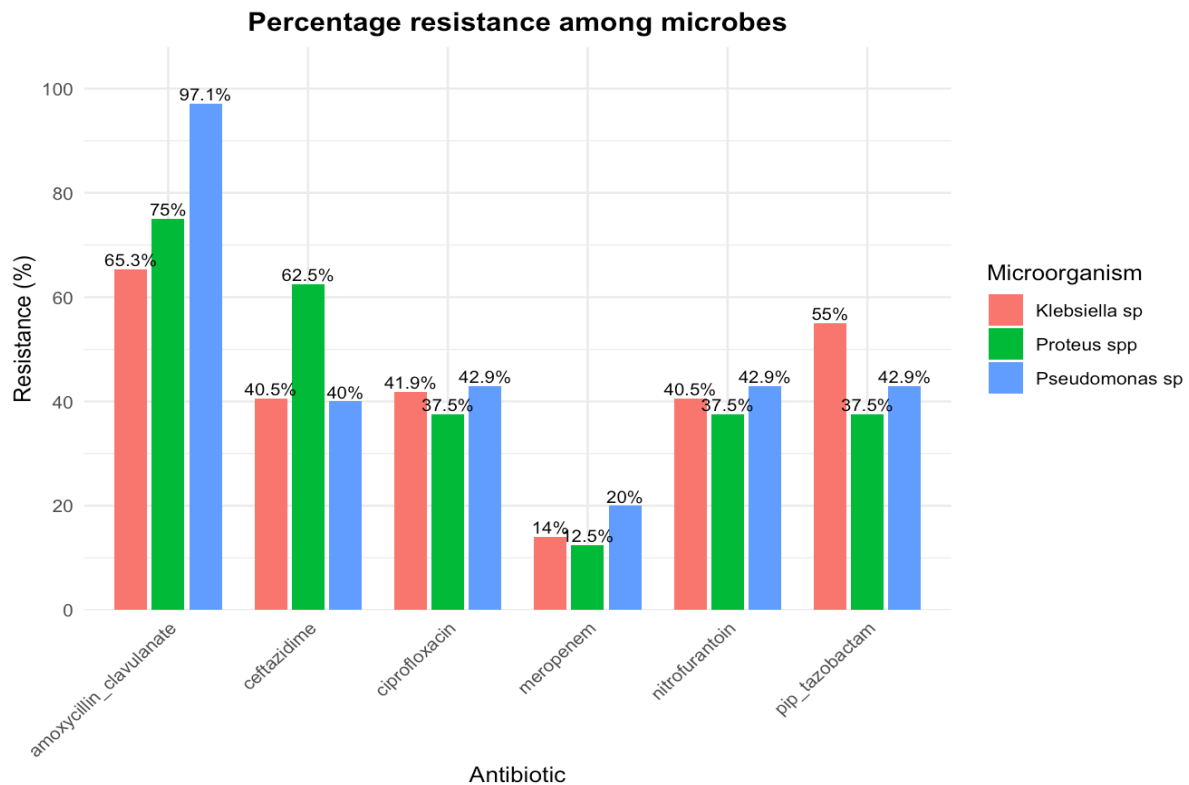


Figure 4: Percentage resistance among Gram-negatives

Resistance among *Klebsiella* sp, *Proteus* spp, and *Pseudomonas* sp was the highest against amoxicillin-clavulanate, with *Pseudomonas* sp resistant in nearly all cases (97.1%), followed by *Klebsiella* sp (65.3%) and *Proteus* spp (75%). Meropenem remained the most effective drug against all three organisms, with resistance not going above 20%. *Proteus* spp showed the highest resistance to ceftazidime (62.5%), and *Klebsiella* sp showed the strongest resistance to piperacillin-tazobactam (55%). Overall, these results suggest many of the antibiotics commonly used in treating infections caused by these pathogens are becoming less effective, while meropenem still remains effective and should be used rationally.

4.3.1 Susceptibility of Ceftriaxone among *E.coli* isolates

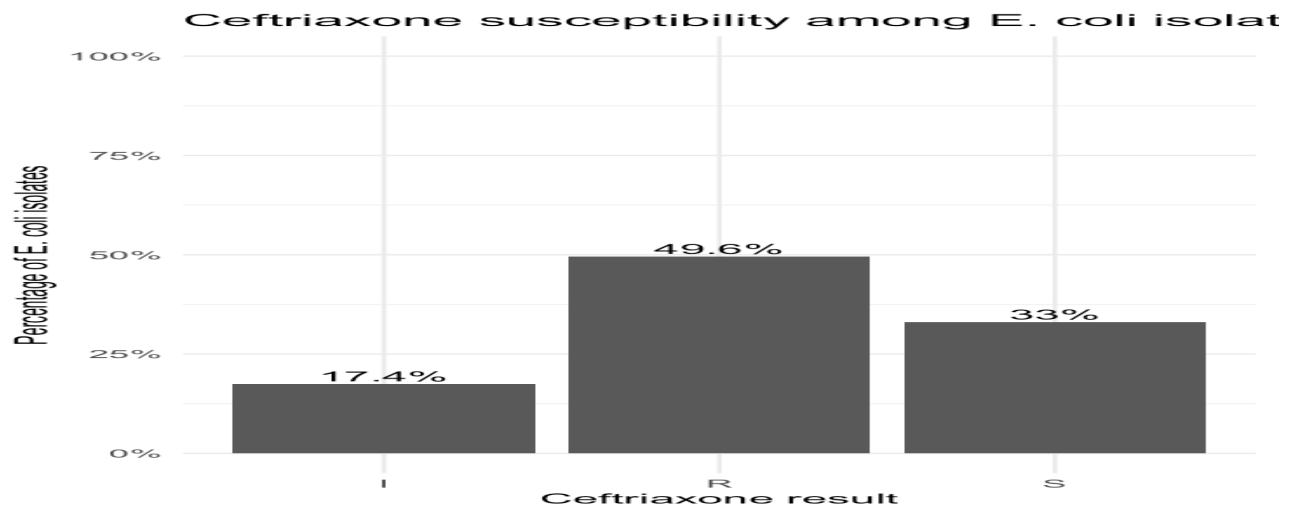


Figure 4: Resistance percentage by antibiotic

Among the *E.coli* isolates that were tested against only ceftriaxone, there was a high level of resistance (49.6%). This is concerning, as ceftriaxone is a third-generation cephalosporin with broad-spectrum activity against many Gram-negative pathogens and is widely used for treating serious infections caused by pathogens such as *E.coli* and *Klebsiella pneumoniae* (Paterson, 2005; Wombwell & Rosa, 2024).

4.3.2. Multidrug Resistance (MDR) among isolates

Table 2: Prevalence of multidrug resistance among isolates

MDR_STATUS	n(%)
Non-MDR	336 (55.6%)
MDR	268(44.4%)

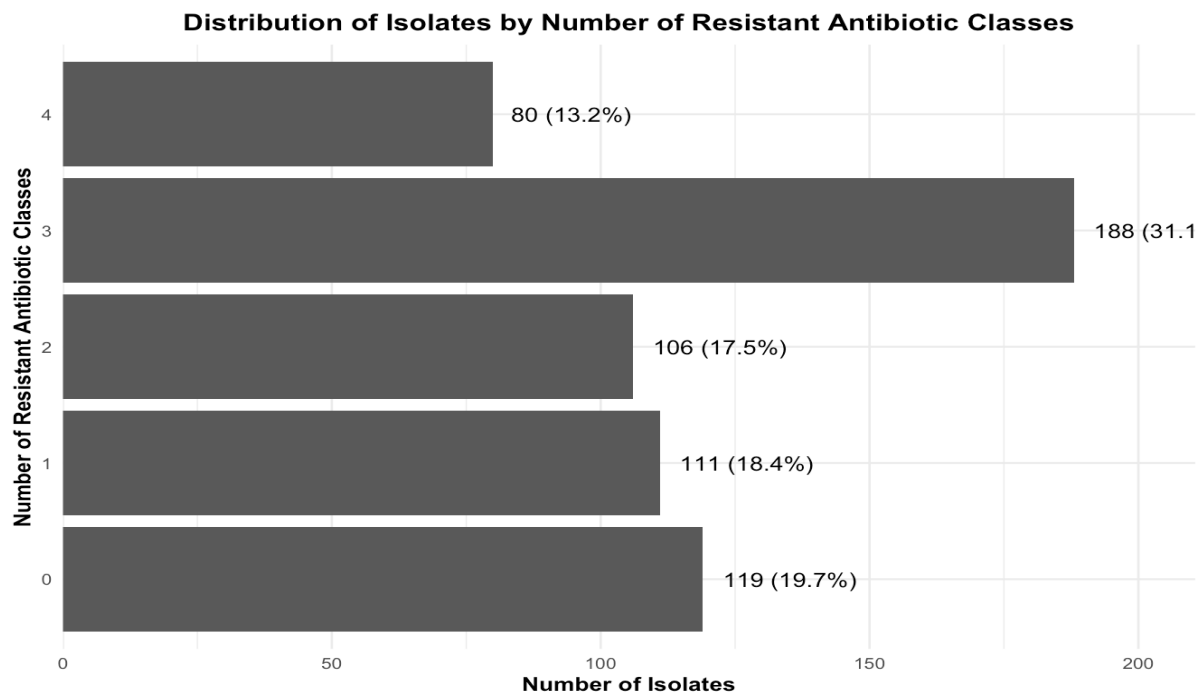


Figure 5: Distribution of isolates by number of resistant antibiotic classes

The prevalence of multidrug-resistant (MDR) isolates was 268 (44.4%), which indicates that nearly half of the isolates exhibited multidrug resistance. This high prevalence poses a significant health concern, as it suggests limited treatment options for infections caused by the microbes (Figure 4). The isolates were resistant to varying numbers of antibiotic classes, with the largest proportion resistant to three classes (188, 31.1%), while 119 isolates (19.7%) showed no resistance (Figure 5).

References

Mareş, C., Petca, R.-C., Popescu, R.-I., Petca, A., Geavlete, B. F., & Jinga, V. (2023).

Uropathogens' Antibiotic Resistance Evolution in a Female Population: A Sequential Multi-Year Comparative Analysis. *Antibiotics*, 12(6), 948.

<https://doi.org/10.3390/antibiotics12060948>

Otaigbe, I. I., Ebeigbe, E., Okunbor, H. N., Oluwole, T. O., & Elikwu, C. J. (2023). Antibiotic susceptibility profiles of Gram-negative bacterial uropathogens in a tertiary hospital,

southwest Nigeria. *African Journal of Clinical and Experimental Microbiology*, 24(3), 299–304.

<https://doi.org/10.4314/ajcem.v24i3.10>

Paterson: *Extended-spectrum β -lactamases: A clinical update*—Google Scholar. (n.d.). Retrieved 24

November 2024, from

https://scholar.google.com/scholar_lookup?author=D.+L.+Paterson&author=R.+A.+Bonomo+&publication_year=2005&title=Extended-spectrum+beta-lactamases+%3A+a+clinical+update&journal=Clin.+Microbiol.+Rev.&volume=18&pages=657-686

Wombwell, E., & Rosa, A. (2024). Comparison of Cefazolin and Ceftriaxone Enterobacterales Susceptibilities for Inpatient Treatment of Urinary Tract Infections and Risk of Hospital-onset *Clostridioides difficile* Infection. *Clinical Therapeutics*, 46(6), 451–454.

<https://doi.org/10.1016/j.clinthera.2024.02.011>

APPENDIX

Link to GitHub Repository

https://github.com/ElizabethBaidoo/Elizabeth_Baidoo_Capstone.git

Supplementary Tables

Table A: Antimicrobial Susceptibility Profiles

antibiotic	tested	Susceptible n (%)	Intermediate n (%)	Resistant n (%)
Amoxycillin-clavulanate	265	56 (21.1%)	24 (9.1%)	185 (69.8%)
Ceftazidime	265	124 (46.8%)	32 (12.1%)	109 (41.1%)
Ciprofloxacin	265	122 (46%)	32 (12.1%)	111 (41.9%)
Meropenem	265	217 (81.9%)	9 (3.4%)	39 (14.7%)
Nitrofurantoin	265	134 (50.6%)	23 (8.7%)	108 (40.8%)
Piperacillin-tazobactam	265	103 (38.9%)	22 (8.3%)	140 (52.8%)

Table B: Ward distribution of isolates

organism	ward_clinic	n	percent
<i>Candida spp</i>	In-patient	80	100
<i>E.coli</i>	In-patient	339	100
<i>Enterococcus</i>	In-patient	1	100
<i>Klebsiella sp</i>	Outpatient	222	100
<i>Proteus spp</i>	In-patient	8	100
<i>Pseudomonas sp</i>	In-patient	35	100
<i>Staphylococcus aureus</i>	In-patient	41	100
<i>Staphylococcus epidermidis</i>	In-patient	80	100
<i>Staphylococcus saprophyticus</i>	In-patient	1	100